User Instructions: Annual Report Deduplication Pipeline

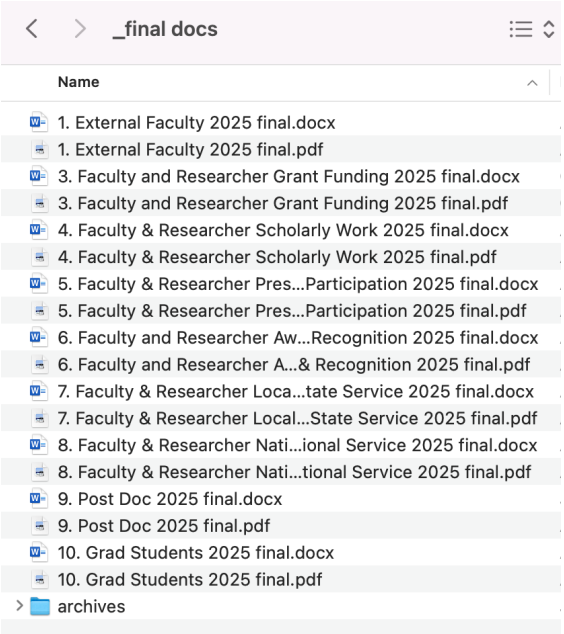
DS6015 Capstone Project – Spring 2026

This guide walks you through preparing your data and running the pipeline to generate accurate, deduplicated publication outputs.

Preparing Your Data

To ensure the pipeline runs correctly, your input data must follow these requirements:

- The pipeline expects a **folder containing all final appendices** from an annual report (e.g., 2025 appendices folder):



- Files within this folder should follow existing naming conventions from past years. Specifically, filenames typically:
  - Contain the word **"final"**, or
  - Contain **" lb"** (with a space),

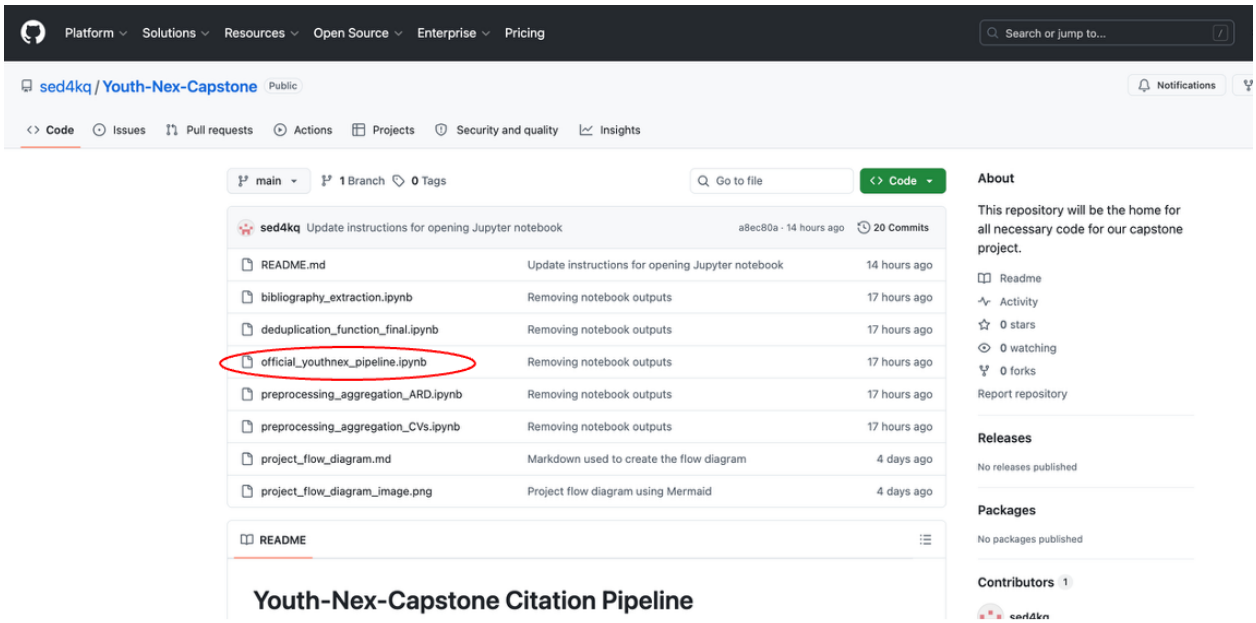
The pipeline is designed to recognize and preserve these conventions.

Once your folder is ready:

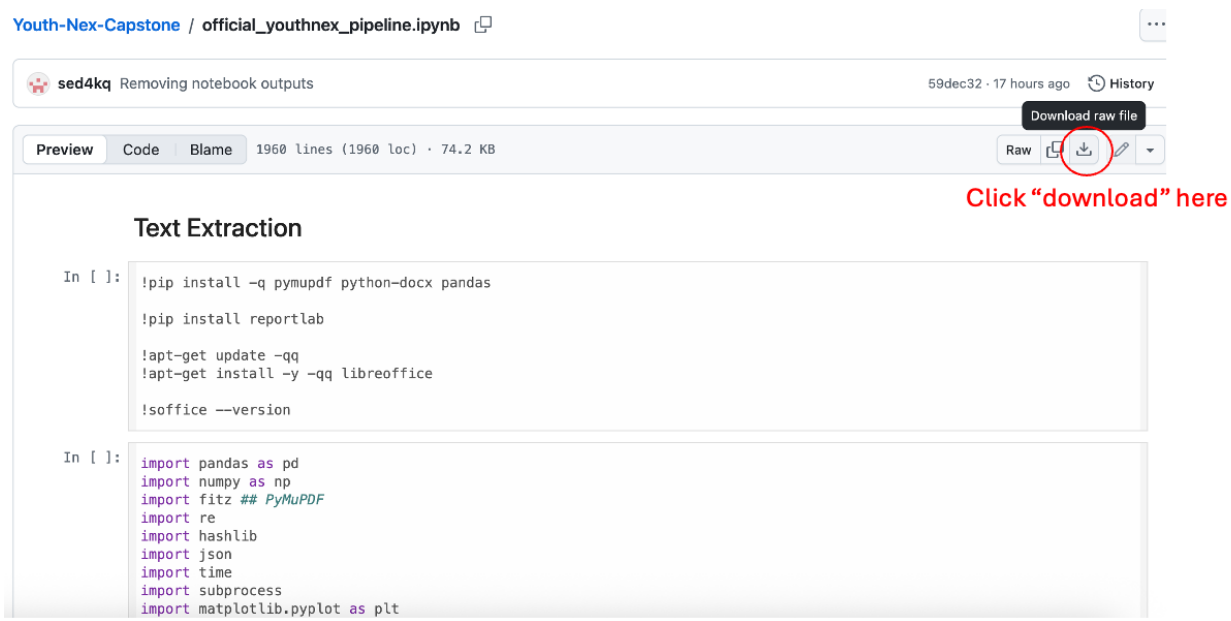
- **Compress the entire folder into a `.zip` file**
- This `.zip` file will be your input for the pipeline

Running the Code

1. Go to the GitHub repository: https://github.com/sed4kq/Youth-Nex-Capstone
2. Locate and select `official_youthnex_pipeline.ipynb` from our repo, where it will direct you to a secondary page.

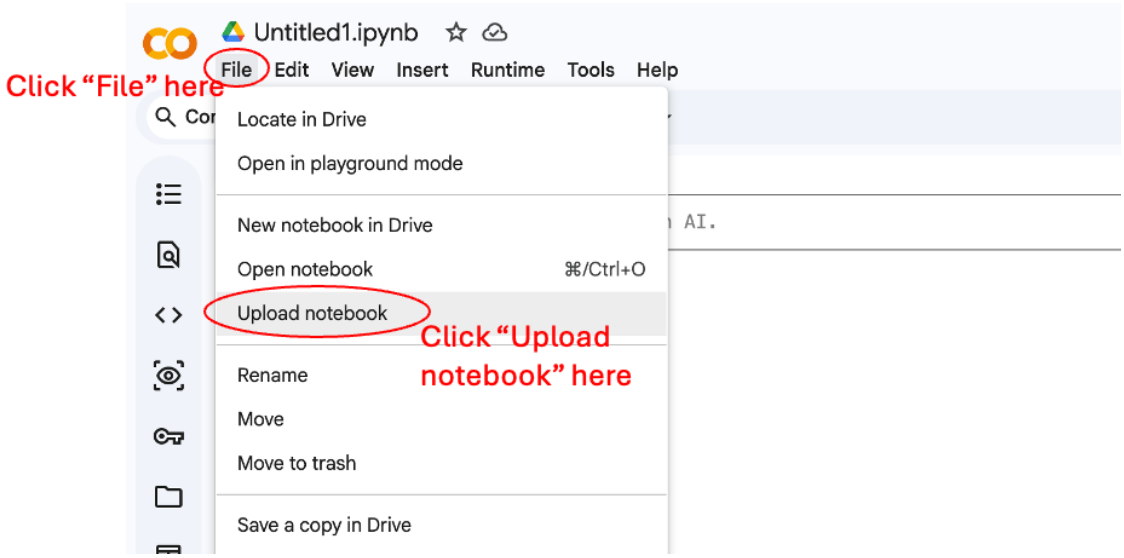


3. Download the notebook to your computer



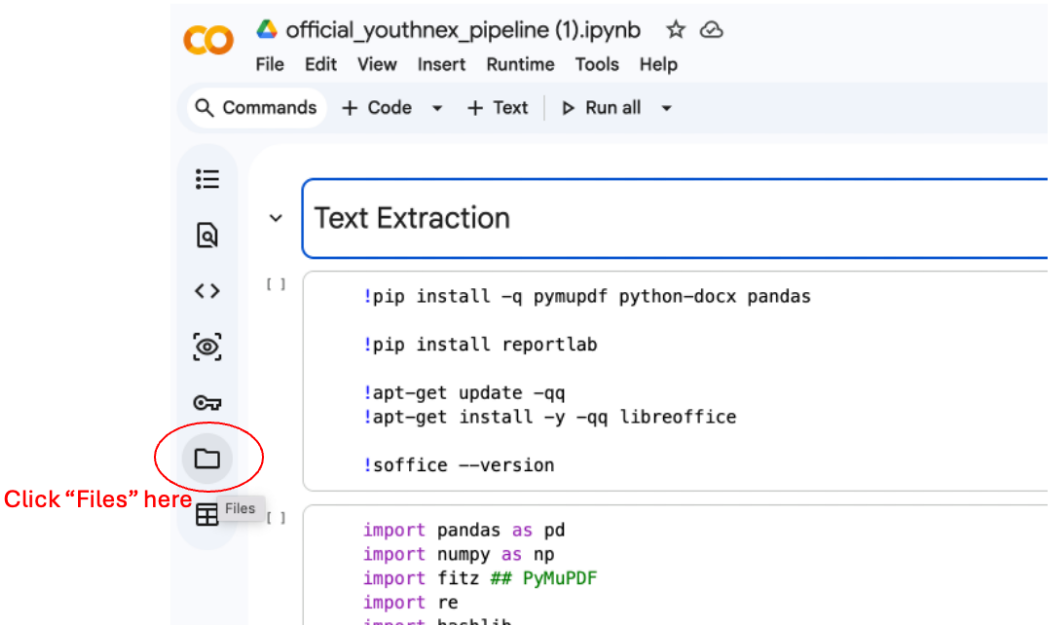
4. Open Google Colab (free with a Google account)

5. Once in Google Colab:
    a. Click **File → Upload notebook**

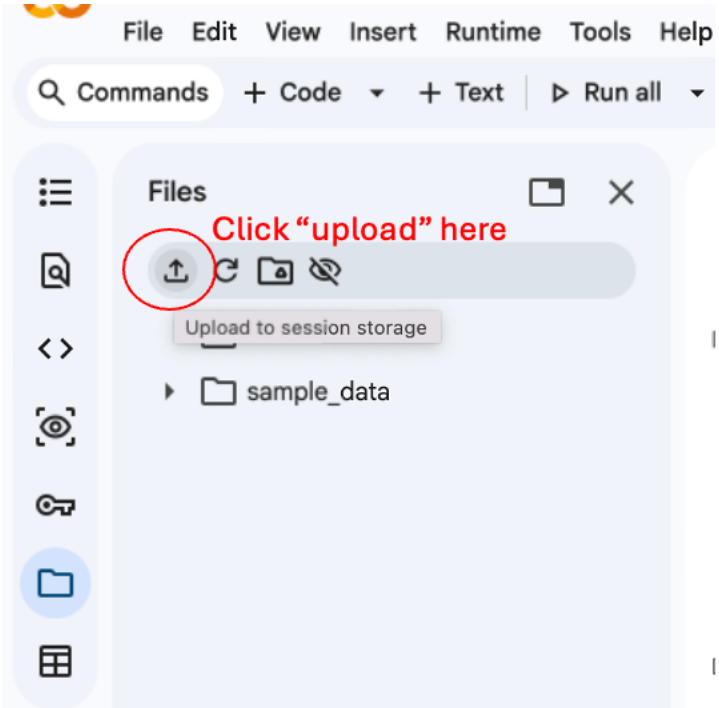


    b. Browse and upload the `official_youthnex_pipeline.ipynb` that you downloaded in Step 3

6. Once the notebook is uploaded:
    a. Click **Files**, displayed as the folder icon on the left sidebar

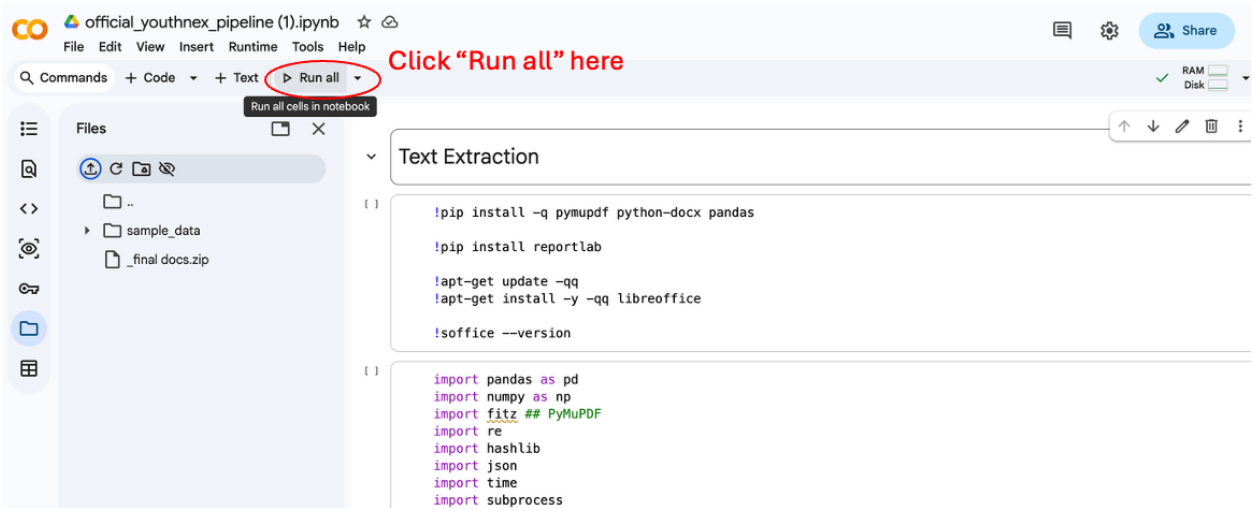


    b. Click **Upload**, displayed as the upload icon



c. Select and upload your `.zip` data file. You may be prompted to select your file twice. You may also get a Warning, select **OK** (this is expected).

7. Once your `.zip` file is uploaded, select **Run All** at the top bar of Google Colab.

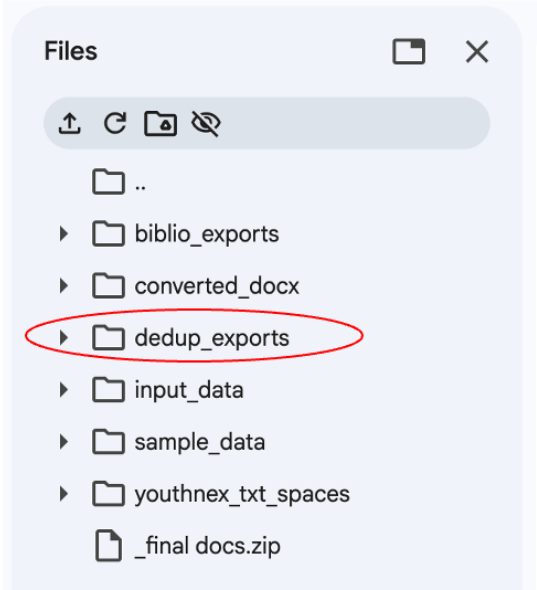


8. Your code should take a few minutes to run.

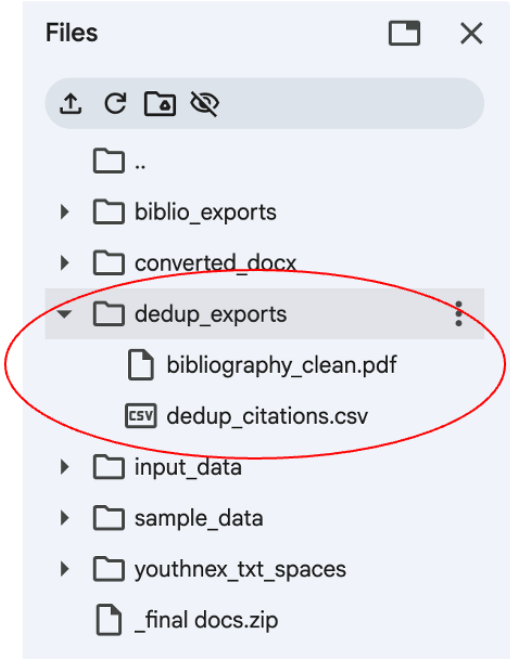
Outputs

Once the code finishes running:

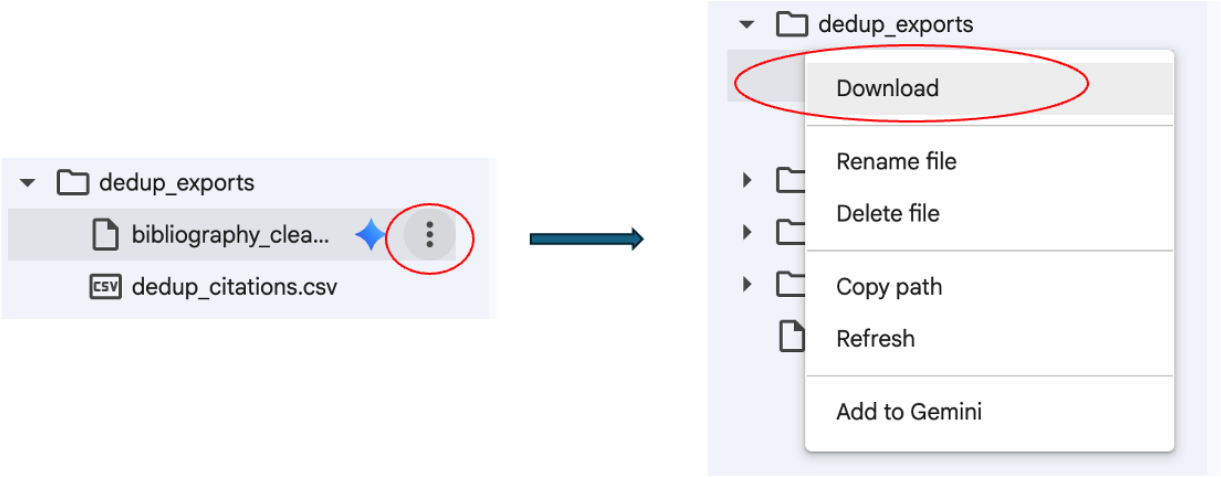
- In the left sidebar, locate the folder `dedup_exports`



- Inside, you will find:
  - `bibliography_clean.pdf`, which is the formatted bibliography output in `.pdf` format
  - `bibliography_clean.csv`, which is the citation dataset in `.csv` format



Download and save these files for your use – you now have an estimated deduplicated bibliography for your reporting, analysis, and visualization needs!



Additional Outputs (Visualizations and Statistics)

In addition to the exported files, the pipeline generates **visualizations and statistics throughout execution** to help summarize and validate results. These may include:

- Unique Citation Numbers
- Reduction Percentage
- Deduplication impact (original vs. unique counts)
- Duplicate group size patterns
- Top 10 Most Frequent Authors
- Most Reused Citations

These visualizations and statistics are displayed directly within Google Colab during runtime and can be:

- **Screenshotted or downloaded manually** for use in reports or presentations
- Used to support **infographics, summaries, or stakeholder reporting**